Data Collection and Preprocessing Phase

Date	11 July 2024
Team ID	SWTID1720011518
Project Title	WCE Curated Colon Disease Classification using Deep Learning
Maximum Marks	6 Marks

Preprocessing

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Section	**Description**
Data Overview	The images have been captured via Wireless Capsule Endoscopy (WCE). There are 3200 training images, 800 testing images and 2000 validation images. All are divided into four categories: Normal, Ulcerative Colitis, Polyps, Esophagitis.
Resizing	Resize images to a specified target size.
Normalization	Normalize pixel values to a specific range.
Data Augmentation	Apply augmentation techniques such as flipping, rotation, shifting, zooming, or shearing.
Denoising	Apply denoising filters to reduce noise in the images.
Edge Detection	Apply edge detection algorithms to highlight prominent edges in the images.

Color Space Conversion	Convert images from one color space to another.
Image Cropping	Crop images to focus on the regions containing objects of interest.
Batch Normalization	Apply batch normalization to the input of each layer in the neural network.
Data Preprocessing Code Screenshots	

Loading Data	`# Define paths`
	`train_path = "/content/train"`
	`test_path = "/content/test"`
	`valid_path = "/content/val"`

Resizing	`# For testing and validation, only rescaling is applied`
	`test_datagen = ImageDataGenerator(rescale=1./255)`
	`valid_datagen = ImageDataGenerator(rescale=1./255)`

Normalization	`train_datagen = ImageDataGenerator(`
	`    rescale=1./255,`
	`    zoom_range=0.2,`
	`    shear_range=0.2,`
	`    preprocessing_function=preprocess_input  # VGG16 preprocessing (mean subtraction)`
	`)`

Data Augmentation	`train_datagen = ImageDataGenerator(`
	`    rescale=1./255,`
	`    zoom_range=0.2,`
	`    shear_range=0.2,`
	`    preprocessing_function=preprocess_input  # VGG16 preprocessing (mean subtraction)`
	`)`

Denoising	`def preprocess_image(img):`
	`    # Denoising with Gaussian blur`
	`    img = cv2.GaussianBlur(img, (5, 5), 0)`

Edge Detection	`# Edge detection with Canny edge detector` `edges = cv2.Canny(gray, 100, 200)`
Color Space Conversion	`# Convert to grayscale` `gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)`

Image Cropping	`# Image cropping (adjust cropping dimensions as needed)` `crop_img = img[50:150, 50:150]`

Batch Normalization	`# Flow data from directories`
	`train_generator = train_datagen.flow_from_directory(`
	`    train_path,`
	`    target_size=(224, 224),`
	`    batch_size=20,`
	`    class_mode='categorical'`

Data Collection and Preprocessing Phase

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Maximum Marks	2 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
WCE Curated Colon diseases.	Sizes and orientation of images were different. Contrast in color of images was different.	Low	Rescaling, Normalization and Gray scaling.
WCE Curated Colon diseases.	Borders of disease patches were hard to too sometimes.	Moderate	Denoising with Gaussian blur and Edge detection with Canny edge detector.

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Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
Project Overview	Healthcare professionals struggle with accurately diagnosing colon diseases due to inconsistent imaging data and complex medical records. Developing a deep learning model to analyze these data sources can enhance diagnostic precision, support early detection, and improve treatment planning and patient outcomes.
Data Collection Plan	The dataset which is used is a public dataset. It is available on Kaggle with the name "Curated Colon Dataset for Deep Learning"
Raw Data Sources Identified	The dataset contains medical imaging data in a curated manner. This initiative intends to accurately categorize diverse on these orders through the analysis of patient data and colonoscopy images therefore facilitating early identification treatment planning and improved patient outcomes

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Kaggle	WCE Curated Colon Disease Dataset Deep Learning	https://www.kaggle.com/datasets/francismon/curated-colon-dataset-for-deep-learning	Images	2 GB	Public